

Thinking Differently About Video Processing at the Edge with AI in Systems Including Professional Displays

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Abstract

With AI enabling the unthought of a decade ago, such as humanoid robotics or large language models, it is easy to overlook that AI can also replace or augment the most traditional computation tasks such as video processing. Video compression innovation seeks to improve the ratio of video quality to bandwidth: three different approaches to using AI target to improve this efficiency. Whilst innovation in video scaling can support video compression efficiency in post-processing it can also recreate detail lost within a scaled down input video sequence. This paper motivates how these innovations can support professional display systems from digital signage to unified communications with contributions including showing how an implementation originally targeted to run on GPU-accelerator cards can run on mobile processors. It is also shown how innovation in compression and scaling for displays can enable thinking differently about the video preparation stage for AI pipelines to provide performance improvement (speed up) and more accurate classification.

Author Keywords

AI; super resolution; CODEC; video processing

1. Introduction

An insatiable audience and creator expectation for higher quality visual experiences helps drive display innovation through higher resolution, frame rates, dynamic range and more. To take advantage of this increasing quality of display it is ever more important to ensure the highest quality is maintained from the video origin (e.g. from RAW form) through to delivery, the techniques to support this are referred to as video processing.

This paper will delve into the fascinating field of the use of AI to replace or augment traditional video processing which is a fundamental part of the computation within display systems. The authors follow the sentiment of [1] in the title quote “the most successful AI right now is what I call boring AI”, that whilst AI-based video processing is not the *sexiest* of AI applications it is practical and has potential for high scale usage.

Whilst there are many video processing functions, video compression and video scaling are the focus of this paper as two of the most computationally intensive operations.

As a domain of focus for the authors, the application of the results to applications in Professional Display Systems will be specifically considered in the paper.

The paper is structured as follows: Section 2 surveys prior work; Target compute types for video processing are discussed in Section 3; Section 4 presents results showing that a super resolution implementation first intended for GPU-accelerator cards can run efficiently on low power AI accelerators such as NPUs or integrated GPUs; The application of video processing

innovation to AI vision pipelines is presented in Section 5; In Section 6 the works are summarized; Thoughts on the greater impact of the research are presented in Section 7.

2. Survey of Prior Work

The use of AI in video processing tasks is not a new one. [2] presents a comprehensive literature survey of the role of AI in the full content delivery pipeline stating applications from video encoding to decoding to the delivery network. From an edge the potential for energy efficiency during transmission (due to lower bit rates) is an interesting observation motivating the value of the AI innovations across different video processing tasks.

2.1. Super Resolution

It is interesting to consider the role of super resolution (SR) across various applications. SR is used with video games for a game engine to render the game play at lower resolution and frame rate with inbuilt GPU technology such as XeSS2 [3] to upscale resolution and frame rate for a high-resolution gaming experience at a lower GPU performance. In [4], Lei and Jiang present the application in social media with interesting insights on use to improve user generated content as cloud-based processing and on device SR upscaling at edge of network. This work leveraged open-source SR code from [5]. This [6] API from Microsoft cites video group calls (in unified communications) as well as social media, improved quality over poor networks and bandwidth optimization to improve content delivery network (CDN) costs as SR applications with stated performance scaling up to 1440p using an NPU. Latest 8K TV specifications [7] show support for SR within the display chipset, which is cited as important to take advantage of the higher display resolution whilst available content is still either FHD or 4K [8]. [9] shows an example of how SR can augment a bi-cubic filter with focus of the AI to add back residuals such as edge sharpness, it also shows broader application to processing including SDR to HDR conversion. The common value propositions across these use cases are to improve quality (adding details that are not present in upscaling such as by bi-cubic) and/or optimizing network bandwidth or compute need. There has so far been an expectation, such as enabling SR in Microsoft Edge [10] requiring adding an Nvidia or AMD GPU card, of the need for acceleration. The results in [6] for up to 1440p and those shown in Section 4 for scaling up to 4K resolution show this need not be the case.

Hallucinations of AI are a common concern when discussing its use for video scaling or CODECs. In [11] an interesting approach to SR with AI is taken to use AI to pre-train filters and includes guard circuitry to help protect against anomalies.

2.2. AI Video CODECs

A fundamental task of video compression is to get the highest quality video signal to the destination at the lowest possible bandwidth, referred to as compression efficiency. AI approaches to video compression, such as [12-15], aim to improve upon this

compression efficiency using the predictive power of AI.

For video CODECs it is interesting to consider 3 roles for AI. First, pre or post-processing video with AI-based approaches for existing CODECs. Second, AI-based approaches to traditional CODEC tools. Thirdly, where the whole video CODEC is rethought to use AI throughout the processing.

As example of the first approach, [14] demonstrated the use of SEI messages with VCC/H.266 to provide a 5% bit-rate reduction using AI-based post-processing with model information sent in the SEI message. In [15], the concept of AI-based pre-processing to implicitly influence CODEC saliency map is another good example of this first type. [16] is a similar example from real-world deployment at the Paris Olympics whereby an AI approach was taken to preset the CODEC to the AI-perceived type of content and best-known settings.

As brief definition, saliency maps are how a CODEC determines where to allocate the most bits in compression to be sympathetic to how the human visual system will view the decoded video.

On the second type, examples include proposals to H.267 [17] or the MPAI-Enhanced Video Coding (EVC) [18] standardization efforts. [15] compares the implicit approach to explicitly guiding the saliency map with an AI-approach as a documented example of how CODECs may be modified with AI techniques.

Examples of the third approach include Deep Render, recently acquired by Interdigital, where testing [12] shows capability to run 1080p video decoding with their AI approach on an Apple Mac Mini M4 38TOP NPU at 69fps with quality higher than AV1 or H.265 in tested bit streams (slightly lower quality than VVC). MPAI-EEV E2E is an example standardization AI-based CODEC project [13]. An interesting observation is that such an AI-based CODEC, like will be shown with the SR testing in Section 4, can potentially be run within modern NPUs such as the upcoming Intel Core Ultra 3[19], leaving the GPU free for graphics or heavy workload AI allowing CODEC innovation to be an NPU software change versus need for silicon changes. [20] shows reduced complexity of AI CODEC for 1080p30 coding, claiming 20x speedup for DCVC-RT versus DCVC-DC/DCVC-FM running on NVIDIA RTX 2080 Ti.

2.3. Learnings for Professional Display Systems

Consider three professional display system related use cases of digital signage, unified communications and a generic supporting AI vision pipeline. From the above prior work the value propositions of SR and AI Video CODECs can be summarized as shown in Table 1.

Table 1. Professional Display Systems Value Proposition

Type	Digital Signage	Unified Comms.	AI Vision Pipelines
Super Resolution	Bandwidth, image detail	Bandwidth, image detail	Quality of results Bandwidth
AI Video CODECs	Lower bandwidth	Lower bandwidth	Bandwidth

A common expected value proposition for both Super Resolution and AI in Video CODECs is to lower bandwidth of transmitted video content. In addition, Super Resolution is expected to be able to recreate image detail not there in the original image, such as cropping small faces within a video feed from a UC camera and enlarging. For this to be practical super resolution must run on target edge compute devices – this will

be explored further in Section 4. For AI vision pipelines, it is expected that super resolution can enhance quality of result, both SR and AI Video CODECs are expected to help lower transmitted compressed video bandwidth– this will be explored further in Section 5.

3. Compute Types for Edge Display Systems

In display systems, video processing is a fundamental requirement in the overall dataflow. [21] presented that there are four types of computer hardware, with (example processor types) which are commonly used to support processing associated with display applications. These being scalar (CPU), vector (GPU), matrix (NPU) and spatial (FPGA or ASIC). It is notable that the vector and spatial types are what historically computed video processing tasks such as video CODECs, scaling and other pre/post-processing. It will be seen later in this paper that matrix types such as neural processing units (NPUs) are destined to become a third common target.

Modern processors today such as Intel Core Ultra 3[19] or Apple M5 [22] include these different compute engine types within a single packaged processor.

Table 2. Processing Type Taxonomy (from [21]) with AI Video Processing Target Updates (column 4)

Type	Example Processor	Display Functions	Updated with AI in video processing
Scalar	CPU	Interactive applications	+ AI video processing (low latency)
Vector	Graphics Processing Unit (GPU)	Rendering, overlay	+AI video processing (highest performance, real-time)
Matrix	Neural Processing Unit (NPU)	Audience measurement	+AI video processing (power efficient, resource balance, analytics)
Spatial	FPGA ASIC	Display controller, LED processor, video splitter	+AI video processing options (specialized video workflows)

The third column of Table 2 shows traditional display functions fitting these processor types. Column 4 provides a summary of where AI based video processing tasks would run based on results from Sections 4 & 5 as well as insights from Section 2.

4. Real-time Edge Performance and Compute Options for Super Resolution

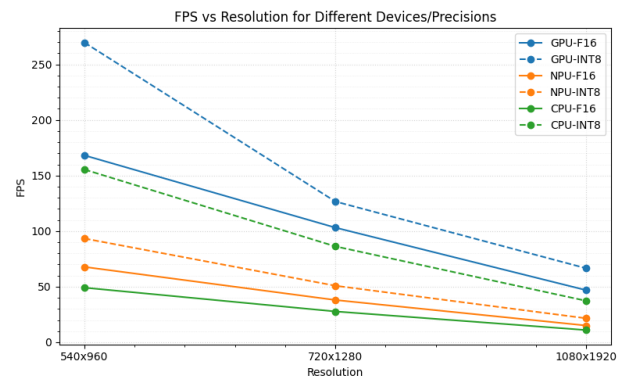


Figure 1: FPS scaling across resolutions for different devices and precisions (target: Intel® Core Ultra 7 Processor 265H, 16GB DDR5 RAM, Ubuntu 24.04 [23]).

In modern on-device AI workloads, understanding how model performance scales with input resolution and numerical precision are crucial for selecting the right hardware and optimization strategy. From the standpoint of system design, the behaviour of a Super Resolution model across devices illustrates a more general principle: the same AI algorithm can occupy very different points in the trade-off space of quality, speed, and hardware constraints. A configuration that comfortably runs on a GPU may, when quantized and paired with a lower-resolution input, still be viable on an embedded NPU or CPU, yet remain good enough for different scenarios. This flexibility suggests that video compression and scaling should not be treated as fixed, but as tuneable components that can be adapted to the capabilities of each deployment target while still improving both perceived visual quality and the effectiveness of downstream AI tasks. The results in Figure 1 demonstrate this solutions space.

Notably at the highest performance, real-time (>60fps) 1080p to 4K scaling is achievable on the integrated GPU at int8 precision. For the NPU (which has 13 TOPS [23] peak performance) this achieves 20fps performance so it can be projected with 50 TOPS on upcoming Intel processors [19] real-time could be possible. Table 2 summarises the learnings from these tests in column 4.

5. Application to AI Vision Pipelines

Figure 2 shows a typical AI Vision Pipeline, concurrent camera streams are decoded, this data needs pre-processing (often downscaling) before object detection, after which further cropping and scaling is needed to prepare objects to be classified as different detections, finally the results are overlayed on the original video content before display. To date the majority focus for performance and accuracy improvement has been on model innovation in the non-shaded boxes. In this section options for innovation in the shaded boxes, by applying the AI-based video processing are considered.

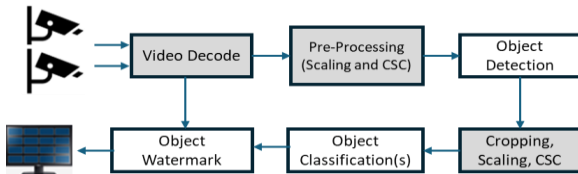


Figure 2: Typical AI Vision Pipeline

[24] presents some foundational work to look at the role of scalable CODECs to improve AI vision pipelines. It is seen that SMPTE VC-6 and MPEG-5 LC-EVC provide compute performance improvements as well as bit stream compression. [25] extends this work to demonstrate power efficiency. Table 3 shows a snapshot of results from [25] with up to 340% speed up over traditional AVC with the LC-EVC scalability. Whilst VC-6 and LC-EVC are not “AI CODECs” these results motivate conceptually that modifying the approach to compression in an AI vision pipeline can have significant performance impact.

Table 3. AI inf. comparison for AVC VS LCEVC-AVC [25]

Video Number	Total fps (8 streams)		Avg per-stream FPS		Fps gain
	AVC	LCEVC AVC	AVC	LCEVC AVC	
Video 1	242.46	824.2	30.31	103.03	340%
Video 2	243.4	692.74	30.42	86.59	285%
Video 3	243.1	821.1	30.39	102.64	338%
Video 4	220.93	693.82	27.62	86.73	314%

One effective way to save bandwidth in such pipelines is to transmit a downsampled video stream and then rely on super-resolution at the receiver. To ensure that detector performance is maintained or even improved on these reconstructed videos, AI-based super-resolution models can be used. Building on this pipeline view, we next ask how changes in the shaded stages, compression and scaling before detection, affect the accuracy of the non-shaded detection model. Low-bitrate compression typically introduces artifacts that harm downstream object detection, so we quantified this effect by measuring mean Average Precision (mAP) with YOLOv11n, a lightweight detector suitable for real-time multi-stream scenarios such as drone body cams. We considered three transmission strategies that correspond to different choices in the “preparation” part of the pipeline: (i) a reference stream at full resolution (1080p) and 1000 kbps (H.265), (ii) a more aggressively compressed 1080p stream at 700 kbps, and (iii) a 540p stream at 700 kbps that is then upsampled back to 1080p using our Super Resolution pipeline before entering the detector. As summarized in Table 4, simply reducing the bitrate from 1000 to 700 kbps at full resolution leads to a measurable drop in mAP of about one percentage point, consistent with the expected impact of compression artifacts. By contrast, the 540p @ 700 kbps + AI Super Resolution configuration recovers essentially the same mAP as the 1000 kbps baseline (and slightly exceeds it on one clip), despite using 30% less bandwidth. These preliminary results suggest that innovation in the shaded boxes of the pipeline, here, AI-based scaling and reconstruction, can partially “undo” the damage of low-bitrate compression and optimize bandwidth usage without compromising detection performance.

Table 4. Effect of Bitrate and AI Super-Resolution on Detection Accuracy (mAP) with VisDrone dataset [26]

Video Number	Mean Average Precision (mAP)		
	1080p @ 1000 kbps	540p @ 700 kbps + AI SR	1080p @ 700 kbps
Video 1	40.30%	40.30%	39.10%
Video 2	15.50%	15.50%	15.30%
Video 3	34.20%	34.40%	33.50%
Video 4	27.00%	27.00%	26.10%

6. Summary

This document explores how artificial intelligence (AI) enhances traditional video processing tasks such as compression and scaling, particularly for edge display systems including professional displays. It highlights the practical benefits of AI-driven video processing in improving video quality, reducing bandwidth, and optimizing AI vision pipelines on various hardware types. Here are some summary findings.

- AI augments video processing tasks: AI can replace or enhance video compression and scaling to improve video quality and bandwidth efficiency, supporting applications like digital signage and unified communications.
- Super resolution applications: AI-based super resolution (SR) improves image detail beyond traditional upscaling, supports bandwidth optimization, and can run efficiently on edge devices without requiring high-end GPU accelerators.
- AI-based video codecs: Three AI codec approaches exist: AI post-processing with traditional codecs, AI-enhanced traditional codec tools, and fully AI-based codecs running on neural processing units (NPUs), offering improved compression

efficiency and quality.

- Edge compute types for video processing: Video processing leverages scalar (CPU), vector (GPU) and matrix (NPU) AI acceleration in modern processors to enable flexible AI video processing deployments at different optimization points.
- Performance on standard processors: Implementations of super resolution originally designed for GPUs can achieve real-time 1080p to 4K scaling on integrated GPUs and promising performance on NPUs, expanding edge processing capabilities without dedicated GPU accelerators.
- Enhancing AI vision pipelines: AI video processing innovations in compression and scaling stages can reduce bandwidth and improve object detection accuracy in vision pipelines, partially mitigating low-bitrate compression artifacts.
- Super Resolution in AI vision pipelines: Early results demonstrate that super resolution can add back lost detail in high compression (such as down to 700 kbps) to achieve accuracy of result of higher bit rate compression (example at 1000 kbps).

As future work the authors plan to perform a deeper analysis of the benefits of super resolution on AI pipelines. Also to analyse in more detail the performance of AI enabled video CODECs to map out performance capability on current and future processors as well as understand the value of these for AI pipelines.

7. Industry Impact Statement

This paper motivates us to think differently about the role of AI in the most traditional video processing tasks. The research indicates that AI-driven video processing can improve perceptual video quality and resource efficiency, influencing future developments in large format IoT display system applications. The work also motivates that innovation in the video preparation stages of AI pipelines with AI-based video processing shows promise to optimise quality of result and/or video bandwidth requirements.

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